

8 Stretching Effects on Injury Reduction and Health

The triumvirate of reasons for stretching over the decades has been to increase range of motion (ROM), improve performance and decrease the chances for injury. The research is overwhelming that we can increase flexibility with stretching (55, 56). However, the next two traditional reasons for stretching are less clear. We will discuss in Chapter 8 how an acute session of prolonged static stretching might actually impair performance rather than improve it. The contention that stretching reduces injuries is also mired in controversy.

An Australian researcher Rodney Pope is well cited for his work in the area. Pope et al. (69) investigated 1,093 Australian army recruits during 12 weeks of military training and found a significant correlation between dorsiflexion range of motion (ROM) and injury incidence. The injuries they followed were ankle sprains, tibia or foot stress fractures, tibial periostitis, Achilles tendonitis, and anterior tibial component syndrome. Poor flexibility was associated with 2.5 times the risk for such injuries compared to average dorsiflexion ROM and eight times the risk compared to people with a high level of flexibility. However, the stretch training programme had no significant effect on the incidence of these injuries. Do these two statements not conflict with each other? Not entirely! The amount of flexibility is an intrinsic factor (something that you have or don't have), whereas stretching is an extrinsic factor (something that you do). So, in the first Pope study, in general if you had better dorsiflexion ROM then you were less likely to get injured. However, doing a static stretching programme did not change the risks to a significant degree. In a similar later study, Pope et al. (70) involved 1,538 Australian army recruits who performed one static stretch each for six lower leg muscles every second day. The stretch training programme did not produce a clinically worthwhile reduction in lower limb injury risk. Injuries included lower body stress fractures, muscle strains, ligament sprains, periostitis, tendonitis, meniscal lesions, compartment syndromes, and bursitis among others. Thus, neither of Pope's studies showed an extrinsic effect of stretching on injury incidence. Small and colleagues (78) reported similar results in their review. Four randomized clinical trials found that static stretching was ineffective in reducing the incidence of exercise-related injury. One of three controlled clinical trials indicated that static stretching decreased

the incidence of exercise-related injury. However, three of the seven studies reported reductions in musculotendinous and ligament injuries although there were no statistically significant reductions in the all types of injury risk. Small concluded that there was moderate to strong evidence that static stretching does not reduce overall injury rates, but may reduce musculotendinous injuries. Weldon and Hill (87) said they could not make a definitive statement, owing to the paucity of well controlled studies at that time. They went as far to say that the scant evidence prior to 2003 would suggest that pre-exercise stretching may actually increase the risk of injury. However, they did not seem that convinced as they also offered that scientific and clinical evidence indicated that stretching in the post-exercise period (cool down) should increase the energy absorbing capabilities of the musculotendinous unit (MTU) reducing the risk of injury. Shrier (77) in his review also indicated that stretching before an event will not reduce injuries. An interesting retrospective study of 3,669 runners found that as you would expect that there was an increased chance of injury in runners if they weighed more, had a previous injury, competed in races (racing at higher intensities and speeds), trained more than two hours and/or more than 20 km per week (higher volumes of training), but surprisingly also if they stretched before running (75). One might speculate that previously injured runners might stretch more to prevent further injuries and that is why stretching was associated with injuries. However, stretching before running was independent of injury history. Based on the hypothesis that stretch training can increase stretch and pain tolerance (62), the runners, if stretching to maximum intensity or point of maximum discomfort may elongate their muscles or tendons past a safe limit and unknowingly inflict muscle and connective tissue injuries (87). In contrast, a randomized, single-blind, non-supervised (self-reports), controlled trial, with self-reporting of muscle, ligament, and tendon injuries found that distance runners who stretched pre- and post-running for 12 weeks experienced a reduction in injuries (0.66 injuries per person-year with stretching versus 0.88 injuries per person-year with controls) (52). Just to confuse the public a bit more, a review by Woods (93) contradicted the earlier reviews by stating that using particular techniques or protocols would provide a reduction in injuries. They indicated that by elongating muscle length to provide a greater ROM led to less musculotendinous unit injuries. They recommended that it was important to ensure an anterior pelvic tilt with the hip at 90° of flexion when stretching the hamstrings. Similarly, a 12-week stretching programme with male youth soccer players, reported increased ROM and a reduction in muscle tightness, which may have contributed to the reduction in non-contact lower limb, trunk, muscle and tendon injuries (6).

The confusion resides in whether the reviews are examining “all-cause” injury risk versus musculotendinous unit injury risk. For example, Amako et al. (4) evaluated military recruits who performed static stretching for 3-months reporting no effect on all-cause injury risk but a significant reduction in musculotendinous injuries, as well as low back pain. How can pre-event

stretching be expected to reduce lower body fractures, meniscal lesions, bursitis or other afflictions? Will stretching reduce overtraining effects typical of long-distance runners and cyclists? It is unlikely according to the literature. But if stretching can improve energy absorbing capabilities of the musculotendinous unit leading to better running economy (lower aerobic output for a given running workload) (38), then you might expect a positive trend for stretching to prevent musculotendinous injuries (66, 90), owing especially to high loads associated with rapid force absorption such as sprinting and agility. Our 2021 review supported that both acute and chronic stretching can reduce the incidence of musculotendinous injuries especially in running-based sports and with explosive actions and change of direction (14). We suggested that this musculotendinous injury protection could be related to stretch-induced changes in the force-length relationship (increased force available at longer muscle lengths) as well as reduced active musculotendinous stiffness, among other factors. As many muscle and tendon injuries occur at extended muscle lengths, greater force at this length could have a protective function. It was also discovered that while the acute effects of stretching on balance is equivocal, chronic stretch training can improve balance, which may contribute to a decreased incidence of injuries associated of slips, trips, and falls (14).

In another one of our reviews (12), Dr. Malachy McHugh (a proud Irishman working in New York) was responsible for the section on injury incidence. In his review of stretching articles investigating injury evidence, he found that eight studies showed some effectiveness of stretching, whereas four showed no effect. There was no evidence that stretching negatively influences (increases) injury risk. Only two of five studies that stretched for less than five minutes indicated reduced injury risk whereas five of six stretching studies with more than five minutes showed an injury incidence benefit for stretching. Only two of five stretching studies showed injury reduction benefits for endurance sports or military training which typically involve overuse injuries. Five of 6 studies showed fewer injuries with sprint running type of sports. The effectiveness of stretching for sprint activities could be related to the high tensile demands placed on the musculotendinous unit with the stretch-shortening cycle. Witvrouw et al. (91) suggest that by increasing the compliance of the tendon, the tissue will have greater energy absorbing capacity. However, as static stretching may not be as effective as ballistic stretching for decreasing tendon stiffness, they recommend including ballistic stretching within the warm-up. Fradkin (36) in another review reported that three of five studies they reviewed showed significant injury prevention benefits with little evidence to suggest stretching was harmful in terms of inflicting injuries. Furthermore, most research (except for one study [22]) reports that stretching does not reduce muscle soreness or other symptoms of muscle damage. Stretching also does not improve recovery from fatigue and may actually inhibit recovery (32). Although, it is a very common practice, one article reports that acute static and proprioceptive neuromuscular facilitation (PNF) stretching did not prevent the frequency of muscle cramping (68). Thus, in summary, pre-

activity stretching of five minutes or more should be more beneficial for injury prevention with sprint running but less effective with endurance type running activities (9)(including military training) that tend to lead to overuse injuries. As workplace injuries are also often overuse injuries, the recommendation to stretch at work may not be as effective as it is often promoted. Behm et al. (12) report the studies in their review indicate a 54 percent risk reduction in acute muscle and tendon injuries associated with stretching.

But, Chapter 8 will emphasize to the reader that more than 60 seconds of static stretching per muscle group without a full warm-up could lead to performance impairments. Important to note that the research says “per muscle group”. To reduce the risk of injury incidence you should stretch more than one muscle group. So, to reduce lower body injuries you might want to static stretch the hip flexors and extensors, knee extensors and flexors, abductors and adductors, dorsiflexors, plantar flexors and ankle evertors for 60 seconds less “each” resulting in perhaps nine minutes of lower body stretching.

Physiological Rationale for Stretch-induced Injury Reduction

Why would stretching have an effect on injury incidence especially with higher speed running activities like sprinting or change of direction (e.g., agility)? Unfortunately, most of the studies that investigate injury risk have not provided a rationale for reduced injuries (66). Theoretically, it is possible that stretching makes the musculotendinous unit more compliant (67, 84), increased compliance shifts the angle–torque relationship to allow greater relative force production at longer muscle lengths (48, 67), and thus the improved resistance to excessive muscle elongation may decrease the susceptibility to muscle strain. Similarly, stretching for 5 minutes can increase tendon compliance and decrease hysteresis (58), which would also allow the tendon to absorb forces over a more extended duration decreasing the stress on this connective tissue minimizing the possibility of a strain. However, a competing rationale could be that increased muscle force output from a lengthened muscle could increase the likelihood of injury. But this rationale would not apply to the risk of other injuries such as ligament sprains, skeletal fractures or overuse injuries. In general, the ability of the MTU to better absorb force perturbations whether it is due to less MTU stiffness (greater compliance) or higher force capabilities at longer muscle lengths should reduce injuries especially with high forces and torques associated with sprinting and agility. Dr. McHugh provided the following theory and application in a personal correspondence. “Muscle strains can occur during concentric contractions and can occur at short muscle lengths. For example, in the days when soccer balls were made with leather and the ball became much heavier in wet conditions it was common to see rectus femoris strains in players kicking a ball (often complete ruptures). At the point of impact with the ball the rectus femoris MTU is shortening (knee is extending, hip is flexing). However, the brain has programmed the body to expect impact with an object with a particular mass.

The actual mass is in fact greater than expected, owing to the wet conditions. At the point of impact with the ball, the contractile and non-contractile elements are shortening synchronously to optimize the impulse imparted on the ball. When the resistance to movement provided by the ball is greater than expected this subtly changes the internal muscle tendon mechanics; at the knee quadriceps and patellar tendon recoil act to extend the knee, if the rate of knee extension is slowed by a heavier than expected ball this will change the loading at the muscle tendon junction (in simplistic terms the tendon is now pulling on the muscle fibres more than the bone because an external force has slowed joint movement). I think that anything that disrupts the interaction between tendon and muscle mechanics during forceful stretch-shortening cycle movements can cause injury to the muscle-tendon unit."

Dr. McHugh continued as follows: "I think the ability of a contracted muscle to absorb sudden tensile loads is important in the mechanism of muscle strains. If an isometrically contracted muscle is stretched suddenly (even a little stretch) there is a large increase in force that is due to the mechanical resistance of the MTU (65, 86). In many cases, owing to the very brief onset, the stretch reflex or other neural changes cannot contribute. I think if the contractile component is able to resist the elongation and impart it to the tendon the muscle will be protected from potential strain injury. Stretching might impact this (albeit in a small way) by allowing greater myofilament overlap at longer muscle lengths than was achievable before the acute bout of stretching, thus allowing the muscle tendon unit elongation to be transferred to the non-contractile component."

While greater compliance will play a role in shock and stress absorption, a musculotendinous unit that is too slack will be hampered in its ability to effectively transfer force from the muscle to the bone to initiate movement. Paradoxically, the muscles and tendons must be compliant and relatively stiff at the same time to prevent injuries while also optimizing performance respectively (14). According to Thorpe et al. (82, 83), the tendon stiffness to muscle strength ratio of individual tendon fibres must be high. However, whereas collagen has the constituency of copper (very stiff and strong), appropriate ROM and compliance can be achieved with efficient sliding between collagen fibres and fascicles. Thus, you can have active stiffness of individual muscle fibres and tendons but you can also have some compliance and an efficient ROM with filament sliding and fascicle rotation (12, 14).

Dynamic Stretching and Activity Effects on Injury Incidence

Our review on the effects of dynamic stretching on injury incidence (11) could only locate two studies that specifically examined dynamic stretching effects in isolation. One study showed no significant effect on injury incidence when comparing a dynamic stretching-only group and a combined dynamic plus static stretching group (95). The second article found increased ankle joint stability with functional dynamic stretching training with injured dancers

(96). However, there was abundant evidence revealing the positive effects of multi-faceted dynamic activity programmes incorporating both dynamic stretching and dynamic activity (i.e., FIFA 11+, FIFA 11 + Kids, FIFA 11 + S, HarmoKnee, Knäkontroll, SISU Idrottsböcker, NMT) (11). These programmes can involve dynamic stretching, jumping, running, bounding, agility, balance, and core stability. There are conflicting reports however on the effects of dynamic stretch training on strength, balance, proprioception, musculotendinous unit stiffness (11). A single session of dynamic stretching should promote thixotropic effects (reduced viscoelasticity owing to stress tension and contraction-induced increases in muscle temperature), contribute to a more positive emotional state, decrease muscle tension, as well as increase concentration, attention, and better prepare athletes for activity and competition (11). With the lack of evidence on dynamic stretching alone, more research is needed on this topic.

Stretching Effects on Posture, Lower Back Pain, and Compensatory Overuse Syndromes

A lack of flexibility in the pelvis, back or other joints can lead to poor posture (23). We strive to keep our centre of gravity over our base of support (equilibrium or balance) or we want to be able to temporarily move our centre of gravity outside our base of support and return to a balanced position (metastability) (54) when reaching or moving. Poor flexibility leading to poor posture as seen with rounded shoulders, or insufficient lumbar spinal curves can hamper our balance or movement capabilities. We then try to over-compensate by engaging other muscles and joints to achieve these functions. The shortened or inflexible muscles may also atrophy and become weaker leading to muscle imbalances (agonist to antagonist muscle activity or prime mover to synergist muscle activity). As the other muscles or joints may not be efficient at achieving these movements, they may be susceptible to fatigue, overuse and perhaps eventual injury. A combination of flexibility training, muscle strengthening and movement re-education might be needed to return the body to proper functioning and reduce the possibility of pain, discomfort and possible injury with these movements.

The role of improving back flexibility to reduce the incidence of low back pain is controversial. It seems that both ends of the spectrum: lack of back flexibility or mobility (19, 24, 85) and excessive spinal mobility (50) can contribute to back problems. However, others have reported that tight hip flexors (33), lack of lumbar or hip motion (33) or thoracic mobility (61) were not related to low back pain. Nonetheless, many authors emphasize the need for an adequate back or trunk mobility in order to ensure proper mechanical functioning (34) otherwise the excessive bending movements on individual vertebrae can lead to an increased injury risk (26). It is suggested by many researchers and organizations that adequate flexibility can reduce the incidence and severity of low back pain (5, 20, 25, 60, 71, 72). Based on the

evidence for and against the protection of lower back problems with greater back ROM (7, 8, 64), Stu McGill suggests that trunk stretching should not emphasize maximum ROM, torso flexibility exercises should be limited to unloaded flexion and extension, ensure adequate hip and knee flexibility so that the back does not have to overcompensate (compensatory relative flexibility [73] for other inflexible joints and finally the first concern should be to improve the strength and endurance of the back prior to incorporating back or trunk flexibility programmes.

Effects of Immobilization on Flexibility

When an individual is injured and subjected to immobilization, the consequences can include muscle atrophy (31, 45, 46), decreases in dynamic (45) and static strength (88, 89), muscle activation (30, 37), reflex potentiation (74), and maximum motor unit firing rates (28, 29) as well as increases in the duration of twitch contractile properties (29, 88, 89). In addition, immobilization can adversely affect flexibility (10). Prolonged immobilization with the muscle in a shortened position can decrease the number of sarcomeres in series, shortening muscle fibre length (39, 79, 80). Fortunately, this sarcomere loss can be rapidly regenerated once normal activity and muscle lengthening occurs (39, 79, 80). Conversely, if a muscle is immobilized in a lengthened position, additional sarcomeres are added (79). Unfortunately, the biomechanics of our joints and muscles dictate that when one muscle is casted in a lengthened position, the other is typically in a shortened position and thus there is no strategy that can benefit both muscles at the same time.

Another factor contributing to a more inflexible muscle with immobilization would be a relative increase in connective tissue, as connective tissue degrades at a slower rate than muscle tissue contributing to a stiffer muscle (79, 80). In addition, there is reported to be a more acute angle of collagen to the axis of the muscle fibres compared to healthy muscle (40) that would also affect stiffness. Thirdly, immobilization reduces the water and glycosaminoglycan content of connective tissue decreasing the distance between collagen fibres resulting in abnormal cross-linkages that limit extensibility of the tissue (2), (63). Stretching tends to counter these deleterious effects (1). However, an immobilized limb in a cast may be difficult to stretch in order to prevent or reduce the decreased extensibility. A counter measure that may be incorporated while the muscle is still immobilized is to stretch the contralateral muscle or actually any other muscle. Unilateral stretching of the hamstrings has improved the ROM of the contralateral hip flexors (21). Furthermore, stretching either the shoulders or the adductor (groin) muscles resulted in improved ROM in the hip flexors and shoulders respectively (13). So, stretch your upper body and your lower body becomes more flexible or stretch your lower body and your upper body becomes more flexible. This is important information for individuals who are immobilized. Just like cross-education (strength increase in contralateral muscle after unilateral training)

(47, 49, 76, 97) is important to preserve strength in an immobilized muscle, stretching unaffected muscles should help to preserve flexibility or at least help to minimize loss of flexibility during immobilization.

Stretching Effects on Dysmenorrhea

Another painful or uncomfortable condition that may be alleviated by stretching is dysmenorrhea (3). Dysmenorrhea is greater than normal pain or discomfort association with menstruation. While it can be attributed in some cases to abnormal oestrogen-progesterone balance; poor posture, owing to shortening of uterine connective tissue which affects the pelvic posterior tilt has also been postulated (41, 71). Static stretching of the pelvis on a consistent basis has been reported to alleviate or reduce dysmenorrhea (41, 42, 43). It is speculated that the stretching decreases the stiffness or restrictions of the fascia and ligaments relieving the compression and irritation of the affected nerves (15, 16).

Stretching Effects on the Cardiovascular System

Stretching can have both positive and negative effects upon the cardiovascular system. One to four minutes of static stretching enhanced blood flow to the stretched muscle for 10 minutes after stretching (57). However, static stretching also acutely increases heart rate, blood pressure and rate pressure product: rate pressure product (RPP) = heart rate (HR) x systolic blood pressure (BP) – a measure of myocardial workload – especially when a Valsalva manoeuvre (exhaling against a closed glottis) was used during the stretching procedure. This increased cardiovascular strain could be dangerous for people with heart and cardiovascular problems (59). However, a meta-analysis revealed that chronic stretch training improved vascular endothelial function, reduced arterial stiffness, resting heart rate, and diastolic blood pressure in middle aged and older adults (53). Another meta-analysis provides further support for the benefits of acute and chronic stretching, reporting significant reductions in arterial stiffness and resting heart rate with no adverse effects on blood pressure with cardiovascular disease patients (81). An additional review indicates that stretch training can improve cardiac autonomic function (increased heart rate variability) in different populations, possibly owing to increases in baroreflex sensitivity, relaxation, and nitric oxide availability (92). Both acute dynamic and static stretching sessions provided large magnitude increases in oxygen saturation, with dynamic stretching showing greater magnitudes (≈ 63 percent) versus static (≈ 55 percent) stretching, which persisted for two minutes after stretching ceased (18). Furthermore, it has been shown that poor flexibility is associated with greater age-related arterial stiffening (94). Remember that associations and correlations are not causative. That means that improving your flexibility will not necessarily decrease arterial stiffness. It may just

mean that people with stiff arteries have not been active throughout their lives and poor flexibility is just one sign that their sedentary lifestyle has contributed to possible arteriosclerosis. However, cyclical stretching can have positive effects on the cell shape, cytoplasmic organization and intracellular processes of vascular smooth muscle cells. When stretching, you are unlikely to substantially stretch the arteries but you can increase the vascular tonus with 1 minute of static stretching (51). As the vasculature cytoskeleton and contractile proteins are associated with and embedded into the extracellular matrix (ECM), stretching can move, deform or place stress on the surrounding ECM and thus stimulate the vascular cell membrane receptors (44). Twelve weeks of passive static stretching improved vascular function and arterial remodelling and arterial stiffness. The stretch-induced enhancements related to central mechanisms returned to baseline within six weeks of detraining (17). Repetitive stretching actions would induce phases of hypoemia and hyperemia, owing to repetitive changes in shearing stress and rate, which could be an effective stimulus to cause positive adaptations in vascular function, arterial stiffness and vascular remodeling (17). So, while we normally think of aerobic exercise and nutrition as being the most important factors for cardiovascular health, stretching can also contribute to a healthy cardiovascular system.

Stretching Effects on Diabetes

The day I was going to submit this second edition, I found this article and thought it should be included. A group from Brazil had diabetic participants (61.5 ± 6.6 years with type II diabetes for 9.4 years) perform eight different static stretches for either 30 or 45 seconds each over eight weeks (35). Both stretching durations showed significant reductions in acute and chronic glycemia. We already know that exercise and activity in general, as well as resistance training, are all beneficial in reducing the need for insulin. Folha et al. (35) demonstrated that the activity (static stretching) can be relatively passive and also reduce type II diabetes associated glycemic problems. How does this occur? My hypothesis is that even passive stretching will cause reflexive-induced contractions of the stretched muscle and probably involve active contractions of the target (agonist), synergist, and antagonist muscles to help move the joint to the full ROM and hold that position. When a muscle is contracted, IGF-1 (insulin-like growth factors) signals the GLUT-4 (glucose) transporters to translocate to the membrane to facilitate glucose entry into the muscle (27). Hence, if the membrane is less sensitive to insulin as with type II diabetes, the activation of GLUT-4 transporters would partially compensate for this insulin sensitivity deficit.

Summary

Generally, there is not strong evidence for stretching programmes to reduce the incidence of all cause injuries. However, there is more convincing, moderate levels of evidence that a static stretch training programme can reduce the incidence of musculotendinous and ligamentous injuries especially with more powerful actions such as sprinting and change of direction (agility) (12, 14) (Figure 8.1). Flexibility training may increase the energy absorbing capability of the musculotendinous unit, as a more compliant tissue will absorb these forces over a more prolonged duration. There is a lack of evidence regarding the effects of dynamic stretching alone on injury incidence but dynamic stretching combined with dynamic activity does seem to produce positive effects on injury prevention. It is unlikely that neural reflexes contribute greatly to this improved protection as the unexpected loads or torques from a slip, fall or collision may be too rapid to be affected by a reflex response. Chronic problems such as low back pain might be positively affected by stretch training as the increased flexibility may improve hip/pelvic orientation affecting spinal curves alleviating back pain and compensatory overuse syndromes. Similarly, stretching may decrease the discomfort of dysmenorrhea by reducing fascia and ligament stiffness and restrictions. It is important to stretch an immobilized limb as there can be a shortening of the muscle, owing primarily to abnormal cross linkages and in very prolonged cases decreases in sarcomeres in series. Even stretching another muscle such as the contralateral homologous or heterologous muscle can help maintain a certain degree of ROM in the affected muscle. While enhanced flexibility is correlated with decreased arterial stiffness and stretching can increase blood

Injury Risk Little evidence that SS decreases all-cause injury risk. Stronger evidence for a SS-induced reduction in MTU injuries, particularly in running-based sports. Ballistic stretching and active warm-up speculated to reduce tendon injuries in SSC activities <u>Speculative Mechanisms</u> Altered muscle force length relationship More compliant MTU Develop appropriate tendon stiffness to muscle strength ratio Improvements in balance	 
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Figure 8.1 Injury risk factors

flow to the stretched muscle, individuals with cardiovascular diseases must be cautious, as a result of stretch-induced increases in the rate pressure product (heart rate x blood pressure: a measure of cardiac workload or demand).

References

1. Ahtikoski, A.M., Koskinen, S.O., Virtanen, P., Kovanen, V., and Takala, T.E.Regulation of synthesis of fibrillar collagens in rat skeletal muscle during immobilization in shortened and lengthened positions. *Acta Physiol Scand* 172: 131–140, 2001.
2. Akeson, W.H., Amiel, D., and Woo, S.Immobility effects on the synovial joints: The pathomechanics of joint contraction. *Biorheology* 1/2: 95–110, 1980.
3. Alter, M.J. *Science of Flexibility*. Champaign, IL: Human Kinetics, 1996.
4. Amako, M., Oda, T., Masuoka, K., Yokoi, H., and Campisi, P.Effect of static stretching on prevention of injuries for military recruits. *Mil Med* 168: 442–446, 2003.
5. American College of Sports Medicine. ACSM's Guidelines for Exercise Testing and Prescription. *Medicine Science Sports and Exercise* 6: 158–164, 2000.
6. Azuma, N. and Someya, F.Injury prevention effects of stretching exercise intervention by physical therapists in male high school soccer players. *Scand J Med Sci Sports* 30: 2178–2192, 2020.
7. Battie, M.C., Biggos, S.J., Fisher, L.D., Spengler, D.M., Hansson, T.H., Nachemson, A.L., and Wortley, M.D.The role of spinal flexibility in back pain complaints within industry: A prospective study. *Spine* 8: 768–773, 1990.
8. Battie, M.C., Biggos, S.J., Sheely, A., and Wortley, M.D.Spinal flexibility and individual factors that influence it. *Physical Therapy*: 67, 1987.
9. Baxter, C., McNaughton, L.R., Sparks, A., Norton, L., and Bentley, D.Impact of stretching on the performance and injury risk of long-distance runners. *Res Sports Med* 25: 78–90, 2017.
10. Behm, D.G.Debilitation to adaptation. *Journal of Strength and Conditioning Research* 7: 65–75, 1993.
11. Behm, D.G., Alizadeh, S., Daneshjoo, A., and Konrad, A.Potential Effects of Dynamic Stretching on Injury Incidence of Athletes: A Narrative Review of Risk Factors. *Sports Med* 53: 1359–1373, 2023.
12. Behm, D.G., Blazeovich, A.J., Kay, A.D., and McHugh, M.Acute effects of muscle stretching on physical performance, range of motion, and injury incidence in healthy active individuals: a systematic review. *Appl Physiol Nutr Metab* 41: 1–11, 2016.
13. Behm, D.G., Cavanaugh, T., Quigley, P., Reid, J.C., Nardi, P.S., and Marchetti, P. H.Acute bouts of upper and lower body static and dynamic stretching increase non-local joint range of motion. *Eur J Appl Physiol* 116: 241–249, 2016.
14. Behm, D.G., Kay, A.D., Trajano, G.S., Alizadeh, S., and Blazeovich, A.J.Effects of stretching on injury risk reduction and balance. *Journal of Clinical Exercise Physiology* 10: 106–116, 2021.
15. Billing, H.E.Dysmenorrhea: The result of a postural defect. *Archive of Surgery* 5: 611–613, 1943.
16. Billing, H.E.Fascial Stretching. *Journal of Physical and Mental Rehabilitation* 1: 4–8, 1951.
17. Bisconti, A.V., Ce, E., Longo, S., Venturelli, M., Coratella, G., Limonta, E., Doria, C., Rampichini, S., and Esposito, F.Evidence for improved systemic and

- local vascular function after long-term passive static stretching training of the musculoskeletal system. *J Physiol* 598: 3645–3666, 2020.
18. Brodeur, Z.R., Paustian, M.J., Monteleone-Haught, D.A., Lamm, R.A., Pagano, A.G., and Ellis, C.E. The Effects of Static and Dynamic Stretching on Muscle Oxygen Saturation in the Rectus Femoris. *Int J Exerc Sci* 15: 702–708, 2022.
 19. Burton, A.K., Tillotson, K.M., and Troup, J.D. Variation in lumbar sagittal mobility with low-back. *Spine* 6: 584–590, 1989.
 20. Cailliet, R. *Low back pain syndrome*. Philadelphia, PA: F.A. Davis, 1988.
 21. Chaouachi, A., Padulo, J., Kasmi, S., Othmen, A.B., Chatra, M., and Behm, D.G. Unilateral static and dynamic hamstrings stretching increases contralateral hip flexion range of motion. *Clin Physiol Funct Imaging* 37: 23–29, 2017.
 22. Chen, H.M., Wang, H.H., Chen, C.H., and Hu, H.M. Effectiveness of a stretching exercise program on low back pain and exercise self-efficacy among nurses in Taiwan: a randomized clinical trial. *Pain Manag Nurs* 15: 283–291, 2014.
 23. Corbin, C.B. and Noble, L. Flexibility: A major component of physical fitness. *Journal of physical Education and Recreation* 6: 23–24, 57–60, 1980.
 24. Curtis, L., Mayer, T.G., and Gatchel, R.J. Physical progress and residual impairment quantification after functional restoration. Part III: Isokinetic and isoinertial lifting capacity. *Spine (Phila Pa 1976)* 19: 401–405, 1994.
 25. Deyo, R.A., Walsh, N.E., Martin, D.C., Schoenfeld, L.S., and Ramamurthy, S.A. controlled trial of transcutaneous electrical nerve stimulation (TENS) and exercise for chronic low back pain. *N Engl J Med* 322: 1627–1634, 1990.
 26. Dolan, P. and Adams, M.A. Influence of lumbar and hip mobility on the bending stresses acting on the lumbar spine. *Clin Biomech (Bristol, Avon)* 8: 185–192, 1993.
 27. Douen, A.G., Ramlal, T., Klip, A., Young, D.A., Cartee, G.D., and Holloszy, J.O. Exercise-induced increase in glucose transporters in plasma membranes of rat skeletal muscle. *Endocrinology* 124: 449–454, 1989.
 28. Duchateau, J. and Hainaut, K. Electrical and mechanical changes in immobilized human muscle. *Journal of Applied Physiology* 62: 2168–2173, 1987.
 29. Duchateau, J. and Hainaut, K. Effects of immobilization on contractile properties recruitment and firing rates of human motor units. *Journal of Physiology* 422: 55–65, 1990.
 30. Duchateau, J. and Hainaut, K. Effects of immobilization on electromyogram power spectrum changes during fatigue. *European Journal of Applied Physiology and Occupational Physiology* 63: 458–462, 1991.
 31. Dudley, G.A., Duvoisin, M.R., Adams, G.R., Meyer, R.A., Belew, A.H., and Buchanan, P. Adaptations to unilateral lower limb suspensions in humans. *Aviation, Space, and Environmental Medicine* 63: 678–683, 1992.
 32. Eguchi, Y., Jinde, M., Murooka, K., Konno, Y., Ohta, M., and Yamato, H. Stretching versus transitory icing: which is the more effective treatment for attenuating muscle fatigue after repeated manual labor? *Eur J Appl Physiol* 114: 2617–2623, 2014.
 33. Esola, M.A., McClure, P.W., Fitzgerald, G.K., and Siegler, S. Analysis of lumbar spine and hip motion during forward bending in subjects with and without a history of low back pain. *Spine (Phila Pa 1976)* 21: 71–78, 1996.
 34. Farfan, H.F. The biomechanical advantage of lordosis and hip extension for upright activity. Man as compared with other anthropoids. *Spine (Phila Pa 1976)* 3: 336–342, 1978.

35. Folha, P.N., Bacurau, R.P.F., Aoki, M.S., and Massa, M. Effect of active static stretching on glycemia in type 2 diabetes. *Manual Therapy, Posturology and Rehabilitation Journal* 22: 1–16, 2024.
36. Fradkin, A.J., Gabbe, B.J., and Cameron, P.A. Does warming up prevent injury in sport? The evidence from randomised controlled trials? *J Sci Med Sport* 9: 214–220, 2006.
37. Fuglsang-Fredriksen, A. and Scheel, U. Transient decrease in number of motor units after immobilisation in man. *Journal of Neurology, Neurosurgery, and Psychiatry* 41: 924–929, 1978.
38. Godges, J.J., MacRae, H., Longdon, C., and Tinberg, C. The effects of two stretching procedures on the economy of walking and jogging. *Journal of Orthopaedics and Sport Physical Therapy* 7: 350–357, 1989.
39. Goldspink, G., Tabary, C., Tabary, J.C., Tardieu, C., and Tardieu, G. Effect of denervation on the adaptation of sarcomere number and muscle extensibility to the functional length of the muscle. *J Physiol* 236: 733–742, 1974.
40. Goldspink, G. and Williams, P.E. The nature of the increased passive resistance in muscle following immobilization of the mouse soleus muscle [proceedings]. *J Physiol* 289: 55P, 1979.
41. Golub, L.J. Exercises that alleviate primary dysmenorrhea. *Contemporary Ob/Gyn* 5: 51–59, 1987.
42. Golub, L.J., Menduke, H., and Lang, W.R. Exercise and dysmenorrhea in young teenagers: A 3-year study. *Obstet Gynecol* 32: 508–511, 1968.
43. Golub, L.J., Lang, W.R., and Menduke, H. Dysmenorrhea in high school and college girls: Relationship to sports participation. *Western Journal of Surgery, Obstetrics and Gynecology* 3: 163–165, 1958.
44. Halka, A.T., Turner, N.J., Carter, A., Ghosh, J., Murphy, M.O., Kirton, J.P., Kielty, C.M., and Walker, M.G. The effects of stretch on vascular smooth muscle cell phenotype in vitro. *Cardiovasc Pathol* 17: 98–102, 2008.
45. Halkjaer-Kristensen, J. and Ingemann-Hansen, T. Wasting of the human quadriceps muscle after knee ligament injuries. IV Dynamic and static muscle function. *Scand J Rehab Med Suppl* 13: 29–37, 1985.
46. Halkjaer-Kristensen, J. and Ingemann-Hansen, T. Wasting of the human quadriceps muscle after ligament injuries. I Anthropometrical consequences. *Scand J Rehab Med Suppl* 13: 5–10, 1985.
47. Hellebrandt, F.A. Cross education: Ipsilateral and contralateral effects of unimanual training. *JAP* 4: 136–144, 1951.
48. Herda, T.J., Cramer, J.T., Ryan, E.D., McHugh, M.P., and Stout, J.R. Acute effects of static versus dynamic stretching on isometric peak torque, electromyography, and mechanomyography of the biceps femoris muscle. *J Strength Cond Res* 22: 809–817, 2008.
49. Hortobagyi, T. Cross Education and the Human Central Nervous System: Mechanisms of Unilateral Interventions Producing Contralateral Adaptations. *IEEE Engineering in Medicine and Biology* January/February: 22–28, 2005.
50. Howes, R.G. and Isdale, I.C. The non-myotendinous force transmission. *Rheumatology and Physical Medicine* 2: 72–77, 1971.
51. Inami, T., Baba, R., Nakagaki, A., and Shimizu, T. Acute changes in peripheral vascular tonus and systemic circulation during static stretching. *Res Sports Med* 23: 167–178, 2015.
52. Jamtvedt, G., Herbert, R.D., Flottorp, S., Odgaard-Jensen, J., Havelrud, K., Barratt, A., Mathieu, E., Burls, A., and Oxman, A.D. A pragmatic randomised trial

- of stretching before and after physical activity to prevent injury and soreness. *Br J Sports Med* 44: 1002–1009, 2010.
53. Kato, M., Nihei Green, F., Hotta, K., Tsukamoto, T., Kurita, Y., Kubo, A., and Takagi, H. The Efficacy of Stretching Exercises on Arterial Stiffness in Middle-Aged and Older Adults: A Meta-Analysis of Randomized and Non-Randomized Controlled Trials. *Int J Environ Res Public Health* 17, 2020.
 54. Kibele, A., Granacher, U., Muehlbauer, T., and Behm, D.G. Stable, Unstable and Metastable States of Equilibrium: Definitions and Applications to Human Movement. *J Sports Sci Med* 14: 885–887, 2015.
 55. Konrad, A., Alizadeh, S., Daneshjoo, A., Anvar, S.H., Graham, A., Zahiri, A., Goudini, R., Edwards, C., Scharf, C., and Behm, D.G. Chronic effects of stretching on range of motion with consideration of potential moderating variables: A systematic review with meta-analysis. *J Sport Health Sci*, 2023.
 56. Konrad, A., Nakamura, M., Warneke, K., Danti, O., and Gabriel, A. The contralateral effects of foam rolling on range of motion and muscle performance. *Eur J Appl Physiol* 123: 1167–1178, 2023.
 57. Kruse, N.T. and Scheuermann, B.W. Effect of self-administered stretching on NIRS-measured oxygenation dynamics. *Clin Physiol Funct Imaging* 36: 126–133, 2016.
 58. Kubo, K., Kanehisa, H., and Fukunaga, T. Effects of transient muscle contractions and stretching on the tendon structures in vivo. *Acta Physiol Scand* 175: 157–164, 2002.
 59. Lima, T.P., Farinatti, P.T., Rubini, E.C., Silva, E.B., and Monteiro, W.D. Hemodynamic responses during and after multiple sets of stretching exercises performed with and without the Valsalva maneuver. *Clinics (Sao Paulo)* 70: 333–338, 2015.
 60. Locke, J.C. Stretching away from back pain, injury. *Occup Health Saf* 52: 8–13, 1983.
 61. Lundberg, G. and Gerdle, B. Correlations between joint and spinal mobility, spinal sagittal configuration, segmental mobility, segmental pain, symptoms and disabilities in female homecare personnel. *Scand J Rehabil Med* 32: 124–133, 2000.
 62. Magnusson, S.P., Simonsen, E.B., Aagaard, P., Sorensen, H., and Kjaer, M. A mechanism for altered flexibility in human skeletal muscle. *Journal of Physiology* 497 (Pt 1): 291–298, 1996.
 63. McDonough, A.L. Effects of immobilization and exercise on articular cartilage – A review of the literature. *The Journal of Orthopaedic and Sports Physical Therapy* 1: 2–5, 1981.
 64. McGill, S.M. Stability: From biomechanical concepts to chiropractic practice. *Journal of the Canadian Chiropractic Association* 2: 75–88, 1999.
 65. McHugh, M. and Hogan, D. Effect of knee flexion angle on active joint stiffness. *Acta Physiologica Scandinavica* 180: 249–254, 2004.
 66. McHugh, M.P. and Cosgrave, C.H. To stretch or not to stretch: the role of stretching in injury prevention and performance. *Scand J Med Sci Sports* 20: 169–181, 2010.
 67. McHugh, M.P. and Nesse, M. Effect of stretching on strength loss and pain after eccentric exercise. *Med Sci Sports Exerc* 40: 566–573, 2008.
 68. Miller, K.C., Harsen, J.D., and Long, B.C. Prophylactic stretching does not reduce cramp susceptibility. *Muscle Nerve*, 2017.

69. Pope, R., Herbert, R., and Kirwan, J. Effects of ankle dorsiflexion range and pre-exercise calf muscle stretching on injury risk in army recruits. *Australian Physiotherapy* 44: 165–177, 1998.
70. Pope, R., Herbert, R., Kirwan, J., and Graham, B. A randomized trial of pre-exercise stretching for prevention of lower-limb injury. *Medicine and Science in Sports and Exercise* 32: 271, 2000.
71. Rasch, P.J. and Burke, J. *Kinesiology and applied anatomy*. Philadelphia, PA: Lea & Febiger, 1989.
72. Russell, G.S. and Highland, T.R. *Care of the lower back*. Columbia, MO: Spine, 1990.
73. Sahrmann, S.A. *Diagnosis and treatment of movement impairment syndromes*. St. Louis, MS: Mosby Publishers Inc., 2002.
74. Sale, D.G., McComas, A.J., MacDougall, J.D., and Upton, A.R.M. Neuromuscular adaptation in human thenar muscles following strength training and immobilization. *Journal of applied physiology* 53: 419–424, 1982.
75. Sanfilippo, D., Beaudart, C., Gaillard A., Bornheim, S., Bruyere, O., and Kaux, J. F. What Are the Main Risk Factors for Lower Extremity Running-Related Injuries? A Retrospective Survey Based on 3669 Respondents. *Orthop J Sports Med* 9: 23259671211043444, 2021.
76. Shima, N., Ishida, K., Katayama, K., Morotome, Y., Sato, Y., and Miyamura, M. Cross education of muscular strength during unilateral resistance training and detraining. *European Journal of Applied Physiology* 86: 287–294, 2002.
77. Shrier, I. Does stretching improve performance?: a systematic and critical review of the literature. *Clin J Sport Med* 14: 267–273, 2004.
78. Small, K., McNaughton, L., and Matthews, M. A systematic review into the efficacy of static stretching as part of a warm-up for the prevention of exercise-related injury. *Res Sports Med* 16: 213–231, 2008.
79. Tabary, J.C., Tabary, C., Tardieu, C., Tardieu, G., and Goldspink, G. Physiological and structural changes in the cat's soleus muscle due to immobilization at different lengths by plaster casts. *J Physiol* 224: 231–244, 1972.
80. Tabary, J.C., Tardieu, C., Tabary, C., Lombard, M., Gagnard, L., and Tardieu, G. Readaptation of the number of sarcomeres in series in isolated soleus fibers after neurectomy. *J Physiol (Paris)* 65: Suppl 3:509A, 1972.
81. Thomas, E., Bellafiore, M., Gentile, A., Paoli, A., Palma, A., and Bianco, A. Cardiovascular Responses to Muscle Stretching: A Systematic Review and Meta-analysis. *Int J Sports Med*, 2021.
82. Thorpe, C.T., Godinho, M.S.C., Riley, G.P., Birch, H.L., Clegg, P.D., and Screen, H.R.C. The interfascicular matrix enables fascicle sliding and recovery in tendon, and behaves more elastically in energy storing tendons. *J Mech Behav Biomed Mater* 52: 85–94, 2015.
83. Thorpe, C.T., Udeze, C.P., Birch, H.L., Clegg, P.D., and Screen, H.R. Capacity for sliding between tendon fascicles decreases with ageing in injury prone equine tendons: a possible mechanism for age-related tendinopathy? *Eur Cell Mater* 25: 48–60, 2013.
84. Toft, E., Espersen, G.T., Külund, S., Sinkjær, T., and Hornemann, B.C. Passive tension of the ankle before and after stretching. *The American Journal of Sports Medicine* 17: 489–494, 1989.
85. Waddell, G., Somerville, D., Henderson, I., and Newton, M. Objective clinical evaluation of physical impairment in chronic low back pain. *Spine (Phila Pa 1976)* 17: 617–628, 1992.

86. Webber, S. and Kriellaars, D. Neuromuscular factors contributing to in vivo eccentric moment generation. *J Appl Physiol* (1985) 83: 40–45, 1997.
87. Weldon, S.M. and Hill, R.H. The efficacy of stretching for prevention of exercise-related injury: a systematic review of the literature. *Man Ther* 8: 141–150, 2003.
88. White, M.J. and Davies, C.T.M. The effects of immobilization, after lower leg fracture, on the contractile properties of human triceps surae. *Clinical Science* 66: 277–282, 1984.
89. White, M.J., Davies, C.T.M., and Brooksby, P. The effects of short-term voluntary immobilization on the contractile properties of the human triceps surae. *J Gen Physiol* 69: 21–27, 1984.
90. Witvrouw, E., Mahieu, N., Danneels, L., and McNair, P. Stretching and injury prevention: an obscure relationship. *Sports Med* 34: 443–449, 2004.
91. Witvrouw, E., Mahieu, N., Roosen, P., and McNair, P. The role of stretching in tendon injuries. *Br J Sports Med* 41: 224–226, 2007.
92. Wong, A. and Figueroa, A. Effects of Acute Stretching Exercise and Training on Heart Rate Variability: A Review. *J Strength Cond Res* 35: 1459–1466, 2021.
93. Woods, K., Bishop, P. and Jones, E. Warm-up and stretching in the prevention of muscular injury. *Sports Med* 37: 1089–1099, 2007.
94. Yamamoto, K., Kawano, H., Gando, Y., Iemitsu, M., Murakami, H., Sanada, K., Tanimoto, M., Ohmori, Y., Higuchi, M., Tabata, I., and Miyachi, M. Poor trunk flexibility is associated with arterial stiffening. *Am J Physiol Heart Circ Physiol* 297: H1314–H1318, 2009.
95. Zakaria, A.A., Kinningham, R.B., and Sen, A. Effects of Static and Dynamic Stretching on Injury Prevention in High School Soccer Athletes: A Randomized Trial. *J Sport Rehabil* 24: 229–235, 2015.
96. Zhang, W. and Bai, N. The role of functional dynamic stretching training in dance sports. *Review Brazil Medicine Esporte* 28: 837–839, 2022.
97. Zhou, S. Chronic neural adaptations to unilateral exercise: mechanisms of cross education. *Exerc Sport Sci Rev* 28: 177–184, 2000.